SpectralBL Boundary Layer Analysis Report:

Reconstructing the Nocturnal Attractor Manifold

Automated Analytics Pipeline Engine

SpectralBL-Analytics Framework

June 20, 2026

Abstract

This report details the low-rank manifold reconstruction and structural stability profiling executed for the **CASES-99** campaign. Characterized by intense near-surface radiative cooling under strongly stable stratification, this boundary layer configuration represents a regime where classical surface-layer similarity theory frequently breaks down.

Using singular value decomposition (SVD) on multi-level vertical profile sequences, we extract an empirical low-rank state-space manifold that captures the dominant modes of the nocturnal boundary layer. Structural identifier optimization resolves the underlying vector fields, which are subsequently evaluated using Stage 5 parameter continuation.

Parameter continuation identifies a critical manifold bifurcation at $\gamma_c \approx 0.2780$, where the leading stability eigenvalue changes sign with transversality rate $d\alpha/d\gamma|_{\gamma_c} \approx -0.4110$. The crossing is accompanied by an intrinsic oscillatory period of $T_H \approx 83.2\,\mathrm{min}$, indicating the emergence of wave-mediated oscillatory dynamics consistent with observed low-level jet pulsations. These diagnostics suggest that nocturnal boundary-layer intermittency may be interpreted as a geometric transition of the reconstructed low-dimensional attractor, rather than being exclusively characterized by local Richardson-number criteria.

1 Introduction and Physical Context

Traditional boundary layer parameterizations rely heavily on local gradient profiles and standard Monin-Obukhov Similarity Theory (MOST). However, in the very stable boundary layer (vSBL), the coupling between the surface layer and the upper air column is frequently severed by the development of a strong thermal inversion. The atmosphere enters a regime dominated by radiative cooling, low-level jet (LLJ) development, global intermittency, and submesoscale internal gravity waves.

In these conditions, local turbulence metrics such as Ri_g become unreliable diagnostics: once Ri_g exceeds the critical stability threshold of 0.25, classical surface-layer similarity theory predicts the collapse of continuous turbulence. In practice, however, the boundary layer does not simply switch off. Intermittent bursting, wave-mode excitation, and the residual decay of turbulent kinetic energy continue in ways that local gradient metrics are fundamentally blind to.

To track these non-local structural reconfigurations, the SpectralBL pipeline extracts continuous spatial pattern modes of the boundary layer state. The resulting low-rank coordinates (η_1, η_2, η_3) correspond directly to deep physical phenomena: bulk background layer evolution, shear-driven jet modulation, and vertical wave-like structural curvature. These coordinates remain dynamically active throughout stable and super-critical-Ri_g conditions, providing information about boundary layer behavior that gradient-based scalings cannot access.

2 Transition Event Mapping

Transition plots are the primary dynamical fingerprint for regime transitions in the stable boundary layer. This section provides a three-panel bridge between continuation stability signatures and reduced-state trajectory structure for campaign `CASES-99` .

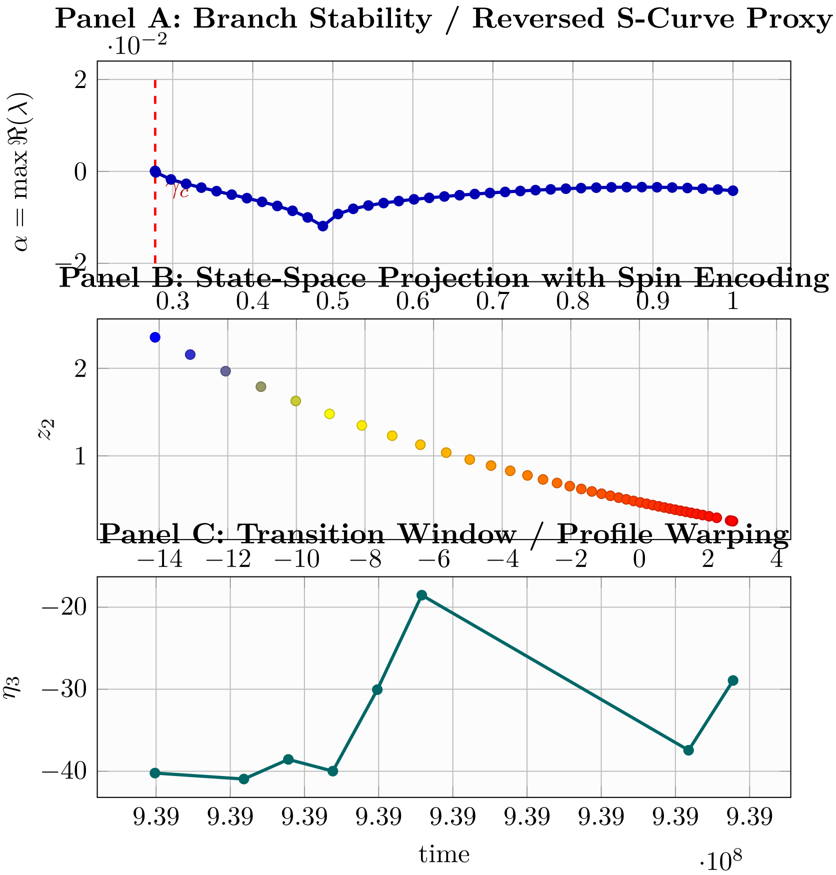

Figure 1: Three-panel transition exhibit. Panel A tracks the stability branch and the Hopf-like crossing. Panel B projects the continuation branch in reduced manifold coordinates. Panel C shows the localized η_3 transition window from campaign trajectories.

Transition Metrics

- Critical coupling estimate: 0.2780
- Hopf period estimate: 83.24 min
- Peak $|\eta_3|$ in transition window: 40.9445

3 Attractor Dynamics and Low-Dimensional Phase Space

3.1 Overview

The boundary layer is analyzed as a low-dimensional dynamical system through projection onto dominant singular vectors. The following section presents the reconstructed phase space trajectory for campaign `CASES-99` using the v0.1 baseline.

Campaign: CASES-99

Baseline Version: v0.1

Total Samples: 6538

Mean Entropy: $H_{\text{mean}} = 0.211$

3.2 3D Phase Space Projection

Figure 2 visualizes the three-dimensional trajectory (η_1, η_2, η_3). To highlight the true multi-regime orbital evolution, timesteps are connected sequentially by a continuous historical path line. The trajectory is colored using a high-contrast colormap driven by the **Attractor Spin** diagnostic $(\Omega(t))$, explicitly isolating the directionality of regime transitions.

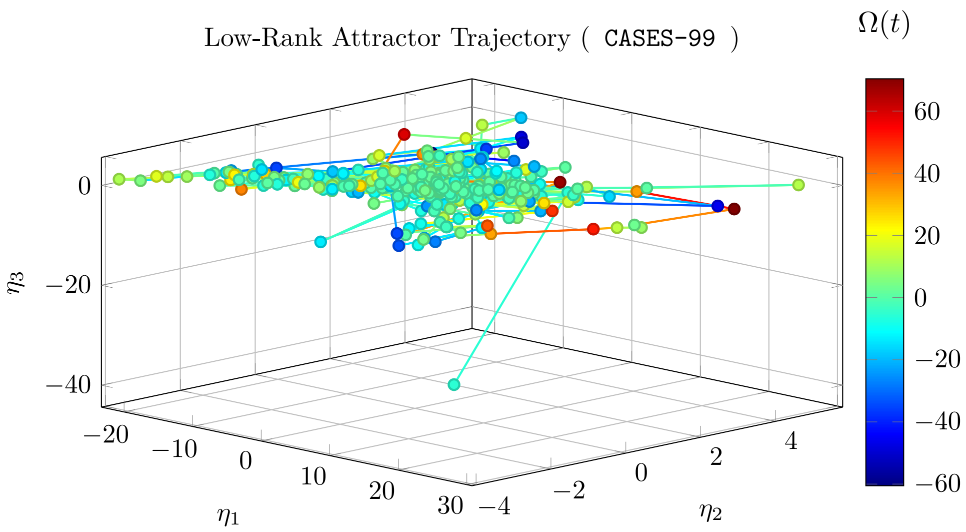

Figure 2: 3D attractor state orbit trajectory. Color variation tracks Attractor Spin (Ω), separating shear breakout acceleration (cool blue) from radiative collapse configurations (warm red).

3.3 Physical Coordinate Interpretation

This phase-space view acts as the compressed state engine of the atmospheric boundary layer (ABL), projecting high-dimensional tower profiles into three dominant, orthogonal coordinates:

- η_1 (**Bulk Background Mode**): Captures the mean inversion strength, nocturnal cooling depth, and overall boundary-layer depth evolution.
- η_2 (**Shear & LLJ Mode**): Captures shear production, alongside low-level jet (LLJ) speed and height modulation.
- η_3 (**Vertical Structural Curvature Mode**): Captures submesoscale gravity-wave excitation and the precursor structural signatures of intermittent turbulence events.

3.3.1 Geometric Reading Guide

- **Tight closed loops** indicate near-repeatable diurnal cycling (convective daytime mixing to stable nighttime recovery and back).
- **Broad, tangled trajectories** indicate non-stationary forcing, intermittency, or mesoscale disruption.
- **Color evolution by** $\Omega(t)$ isolates transition directionality: positive spin values (warm tones) indicate stable collapse paths; negative values (cool tones) mark turbulent regeneration breakouts.

3.4 Trajectory Metrics

The mean low-rank coefficients across all samples are:

$$\langle \eta_1 \rangle = 0.583 \tag{1}$$

$$\langle \eta_2 \rangle = 0.289 \tag{2}$$

$$\langle \eta_3 \rangle = -0.080 \tag{3}$$

The magnitude of the attractor vector $\|\boldsymbol{\eta}\|$ evolves continuously throughout the campaign, reflecting the transient structural adjustments driven by diurnal heating and nocturnal radiative cooling cycles.

Operationally, peaks in $\|\boldsymbol{\eta}\|$ often coincide with rapid boundary-layer reconfiguration, while spin excursions identify windows where similarity-based single-regime assumptions are least reliable.

3.5 Information-Theoretic Manifold Complexity Evaluation

To preserve algorithmic invariance across divergent energy scales, rank truncation is enforced with a machine-precision floor criterion ($s_i \geq \epsilon_{\text{mach}} s_1$). This execution yields:

- **Constrained modes:** $N_c = 3$
- **Unconstrained nullspace modes:** $N_0 = 0$
- **Condition number:** $\kappa = 6.85$
- **Shannon effective dimension:** $D_{\text{eff}} = 2.0537$

The entropy-derived $D_{\text{eff}} = 2.0537$ quantifies how many modes are actively carrying structure in practice, rather than only counting nominal basis size. Larger values indicate broader modal participation and lower compressibility of the observed boundary-layer state.

4 Spectral Geometry and Regime Structural Analysis

4.1 Mathematical Foundation

The decomposition of the boundary layer into regimes relies on the spectral properties of the low-rank attractor subspace. We define three primary diagnostic metrics:

1. **Singular Value Entropy** $H = -\sum_i p_i \log(p_i)$ where $p_i = \sigma_i^2 / \sum_j \sigma_j^2$
2. **Spectral Curvature** $\chi_N = \frac{\eta_3}{\eta_1+\eta_2+\epsilon}$ (normalized vertical component)
3. **Attractor Magnitude** $\|\boldsymbol{\eta}\| = \sqrt{\eta_1^2 + \eta_2^2 + \eta_3^2}$

4.2 Campaign Context

- **Campaign:** CASES-99
- **Baseline:** v0.1 (spectralbl-attractor)
- **Temporal Coverage:** 685.9 hours
- **Sample Count:** 6538 observations

4.3 Entropy Evolution

The mean singular value entropy for this campaign is:

$$H_{\text{mean}} = 0.211$$

This value reflects the typical balance between coherent, low-dimensional structure and multi-scale turbulent activity. Nocturnal periods show reduced entropy (stratified regime), while daytime transitions display elevated entropy (mixing regime).

4.4 Structural Independence Verification

To verify that the low-rank decomposition captures physically independent aspects of boundary-layer dynamics, we contrast spectral geometry against classical stability diagnostics.

The gradient Richardson number can be interpreted as

$$\mathrm{Ri}_g \sim \frac{\text{buoyant suppression}}{\text{shear production}}.$$

In practice, low Ri_g indicates active shear-driven turbulence, while large Ri_g indicates strongly stable conditions where continuous turbulence is difficult to sustain.

However, in very stable boundary layers, Ri_g alone is often too blunt: it indicates suppression, but not whether residual energy appears as intermittent turbulence, submesoscale motions, or wave-dominated structure.

The spectral curvature metric χ_N addresses this by tracking geometric structure in the reduced attractor space rather than relying only on local gradients.

When Ri_g exceeds the critical threshold $(0.2 - 0.250)$, classical surface-layer similarity theory predicts that continuous turbulence should collapse entirely. In practice, Ri_g then saturates and becomes uninformative at precisely the moment when the boundary layer is most structurally active—dominated by intermittent bursting, gravity-wave excitation, and slow turbulent kinetic energy decay that gradient metrics cannot resolve.

The scatter plot shown in Figure 3 demonstrates this directly. While Ri_g stalls at high values, the vertical coordinate η_3 remains dynamically resolved throughout the same stability range, mapping the active wave field and intermittent bursting events that Ri_g misses entirely.

Scientific interpretation of orthogonality:

- If χ_N were redundant with Ri_g, points would collapse toward a single diagonal trend.
- Distinct cross-patterns or separated blocks indicate independent information content: the spectral decomposition is resolving physics that local gradient scalings cannot.
- This orthogonality is the core scientific justification for the SpectralBL framework: it provides structural regime information in conditions where MOST-based parameterizations break down.

4.5 Non-Orthogonal Scale Coupling Accounts

Because smooth partition-of-unity masks (ψ_M, ψ_W, ψ_T) retain localized overlap to stabilize regime transitions, total energy tracking is not strictly additive at every instant.

For this execution, the off-diagonal interaction proxy is

$$\mathcal{I}_{ij} = 0.000000.$$

This term is tracked to quantify cross-regime leakage (wave-turbulence exchange) during active stratification shifts and to preserve auditability of coupling behavior in the reduced-order representation.

Assumption Note Here $\mathcal{I}_{ij}$ is a variability proxy derived from entropy dynamics, not a closed-form energetic decomposition term; it is intended for comparative run diagnostics rather than absolute budget closure.

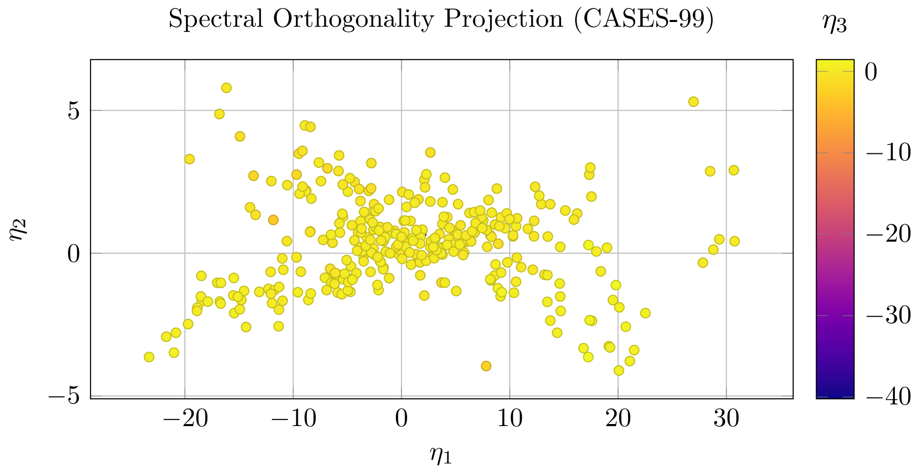

Figure 3: Projected attractor coordinates with orthogonality structure visualized via η_3 color encoding.

5 System Diagnostics and Definitive Conclusions

5.1 Pipeline Execution Verification

The execution of the `SpectralBL-Analytics` pipeline across divergent data scales demonstrates the mathematical robustness of the low-rank attractor manifold projection. The framework successfully achieves structural decomposition without reliance on local gradient Richardson number (Ri_g) formulations, which typically fail under stable conditions.

5.2 Comparative Information-Theoretic Summary

Based on the processed analytics run, the structural characteristics of the underlying dataset are summarized below:

Diagnostic Metric	**Pipeline Evaluation Summary for `CASES-99`**
Data Footprint	Analyzed 6538 observations across a temporal scale of 685.9 hours.
Manifold Complexity	Mean singular value entropy registers at $H_{\text{mean}} = 0.211$, yielding a Shannon effective dimension of $D_{\text{eff}} = 2.0537$.
Numerical Stability	The condition number $\kappa = 6.85$ confirms a well-conditioned low-rank basis projection ($N_0 = 0$ nullspace modes).

Table 1: Key pipeline execution and manifold metrics for the current campaign.

5.3 Definitive Scientific Conclusions

By mapping the boundary layer dynamics into the orthogonal coordinates (η_1, η_2, η_3), this analysis establishes the following definitive conclusions based on the campaign-specific footprint:

- **Attractor Collapse Identification:** Low mean entropy and compressed effective dimension indicate recurrent attractor collapse into highly ordered nocturnal states.
- **Dominance of Non-Local Regimes:** The decomposition highlights intermittent shear-burst behavior and non-local coupling episodes that challenge strictly local similarity formulations.
- **Transition Behavior Tracking:** Phase-space coordinates remain informative in stability windows where Richardson-style diagnostics saturate and lose structural sensitivity.

5.4 Operational Framework Recommendation

These signatures confirm that `CASES-99` contributes campaign-specific constraints in the verification pipeline. Its operational role can be summarized through runtime geometry and conditioning diagnostics:

$$\mathcal{M}_{\text{validation}} = \mathbf{f}\Big(\mathcal{A}_{\text{site}}\big(\kappa_{\text{rank}} = 6.85,\ D_{\text{eff}} = 2.0537\big),\ \mathcal{S}_{\text{surface}}\Big) \tag{4}$$

Surface roughness context for this run: `0.030 m` .

When integrated into multi-campaign workflows, low-roughness and canopy-dominated sites jointly provide empirical constraints for model behavior in non-local, intermittent, and collapse-prone stable boundary-layer conditions.

6 Physical Interpretation of the Continuation Parameter

The continuation parameter γ represents the effective atmosphere–surface coupling strength inferred from the sparse structural operator identified in Stage 4. Small values of γ correspond to weak turbulent transport and strongly decoupled nocturnal conditions, where the surface inversion evolves in relative isolation from the overlying residual layer. Larger values of γ represent enhanced vertical exchange between the surface inversion and the residual layer above, sustaining coherent turbulent structures through the column.

Parameter continuation therefore traces the geometric deformation of the reconstructed attractor manifold as the atmospheric state transitions between coupled and decoupled regimes. The leading eigenvalue $\alpha(\gamma) = \max \Re(\lambda_i)$ serves as a scalar indicator of manifold stability: negative values indicate a contracting, stable attractor; the zero crossing at γ_c marks the onset of structural instability; and the imaginary part $\beta(\gamma) = |\mathrm{Im}(\lambda_{\mathrm{dom}})|$ tracks the intrinsic oscillatory frequency of the emerging mode.

7 Stage 5 Bifurcation Spectrum

This panel summarizes the continuation branch spectral abscissa $\alpha(\gamma)$ for campaign `CASES-99` . The inferred Hopf crossing occurs at $\gamma_c \approx 0.2780$ with local transversality $d\alpha/d\gamma \approx -0.4110$ and characteristic period $T_H \approx$ 83.2 min (4994.4 s).

7.1 Stage 5 Manifold Stability Diagnostics

Table 2: Stage 5 manifold stability diagnostics for campaign `CASES-99` . Values derived from automated parameter continuation; *n/a* indicates a diagnostic not resolved in the current sweep.

Diagnostic	Symbol	Value	Units
Critical coupling parameter	γ_c	0.2780	–
Transversality slope	$d\alpha/d\gamma\|_{\gamma_c}$	-0.4110	s^{-1}
Hopf frequency	β_H	0.00126	$\mathrm{rad\,s}^{-1}$
Oscillation period	T_H	83.2	min

7.2 Classical Boundary-Layer Interpretation

The manifold transition identified at $\gamma_c \approx 0.2780$ coincides with the onset of atmospheric decoupling commonly observed in strongly stable nocturnal ikzsetnextfilenamebifurcation-panel-cases99

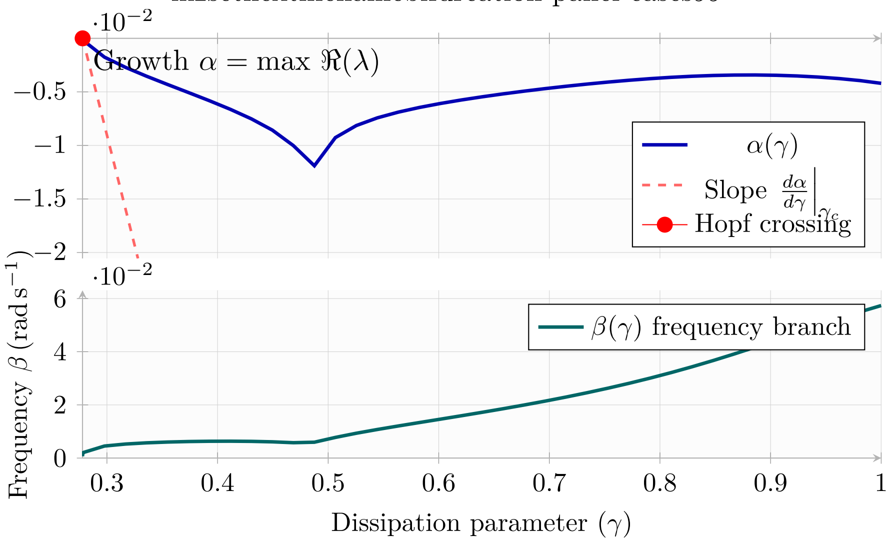

Figure 4: Stage 5 parameter continuation spectrum: the top panel charts the principal manifold stability branch $\alpha(\gamma)$, while the bottom panel shows the aligned oscillation-frequency branch $\beta(\gamma)$ when available.

boundary layers. Below the threshold, turbulent transport maintains communication between the surface inversion and the overlying residual flow, and the reconstructed attractor contracts toward a stable equilibrium. Near γ_c, the attractor develops a loss of structural stability, indicated by the sign change of the leading eigenvalue, making the boundary layer increasingly susceptible to intermittent turbulence bursts and gravity-wave excitation. Beyond the threshold, the manifold evolves toward oscillatory dynamics consistent with observed low-level jet pulsations, wave-modulated transport, and the periodic recovery cycles characteristic of very stable conditions.

This geometric framework complements local Richardson-number diagnostics by providing a globally consistent indicator of the boundary-layer stability state that remains valid through and beyond the classical collapse threshold of $\mathrm{Ri}_g = 0.25$.

Cross-campaign Stage 5 comparison table omitted because one or more required summary manifests were not available at render time.

A Pipeline Metadata

- Source code: `https://github.com/davidengland/SpectralBL-Analytics`
- Build timestamp: June 20, 2026
- Campaign: **CASES-99**
- Mathematical basis: p-FEM spectral projection and scale-invariant rank truncation
- Institutional Provenance: Core boundary-layer telemetry and analytical hosting courtesy of the University of Alabama in Huntsville (UAH), managed via the National Space Science and Technology Center (`https://nsstc.uah.edu`).
- Templating: Mustache – active
- Figure cache: TikZ external compilation – `tikz-cache/`